NEPTUNET

A Lightweight Vision-Based Perception Framework for Underwater Gas Pipeline Inspection

Embedded AI • Underwater Computer Vision • Resource-Aware Perception

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RESEARCH ORIENTED FRAMEWORK

Research Motivation

Conventional subsea pipeline inspection remains costly, intermittent, and difficult to scale over long distances. At the same time, underwater visual perception is challenged by turbidity, scattering, illumination variations, and domain shifts between controlled and real environments. These constraints motivate the development of lightweight embedded perception systems capable of providing inspection-relevant information directly from underwater imagery.

Proposed Framework

NeptuNet is a lightweight hierarchical vision-based perception framework for underwater gas pipeline inspection. The framework organizes perception into three complementary levels: continuous pipeline geometric perception, bubble-based early warning, and conditional leak confirmation. By activating perception modules according to inspection context, NeptuNet aims to balance inspection capability with embedded computational constraints.

HIERARCHICAL PERCEPTION ARCHITECTURE

Level 1. Pipeline Monitoring

Continuous pipeline perception and geometric awareness

- Starts the perception cycle
- Detects and localizes the pipeline
- Estimates orientation, and direction

Level 2. Bubble Analysis

Lightweight early warning from bubble behavior

- Analyzes temporal bubble patterns
- Uses physics-inspired descriptors
- Generates leak-suspicion alerts

Level 3. Leak Confirmation

Suspicion-triggered gas plume analysis

- Triggered by suspicious activity
- Detects and segments gas plumes
- Estimates Plume source and direction

Resource-Aware Logic

NeptuNet activates perception modules according to the inspection context. Pipeline monitoring remains continuously active, bubble analysis provides lightweight early warning, and leak confirmation is performed only when suspicious activity is detected.

Continuous Monitoring → Early Warning → Leak Confirmation

Key Contributions

- **NeptuNet Framework:** Hierarchical perception architecture for underwater pipeline inspection.
- **Pipeline Geometric Perception:** Lightweight segmentation and PCA geometry extraction.
- **Leak Localization:** Gas plume segmentation and source estimation using PDTS.
- **Embedded AI Deployment:** ONNX optimization, INT8 quantization, and NPU benchmarking.
- **Open Research Resources:** Public datasets, reproducible experiments, and research papers.

INNOVATIVE PROJECT LABEL

Awarded by the Ministry of Knowledge Economy, Startups and Micro-enterprises, Algeria.

Recognizes the innovation potential of the NeptuNet framework in underwater AI, embedded perception, and intelligent inspection technologies.

Research Identity

NeptuNet positions underwater inspection as an embedded perception problem rather than a full robotic autonomy problem. The framework combines computer vision, embedded AI, explainable machine learning, and resource-aware system design to support underwater gas pipeline inspection under practical deployment constraints.

CROSS-DOMAIN VALIDATION | PUBLIC DATASETS | DEPLOYMENT OPTIMIZATION | NPU BENCHMARKING | EXPLAINABLE AI

NeptuNet Project Repository
Source code • Datasets • Publications • Resources

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